



Prepared with Synoptic

Final investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine

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Investigation summary

What happened

At 02:47:12 on 12/08/2026, LV-118, a light vehicle, drove out of the RL 312 dewatering spur ahead of HT-407. HT-407 was an empty 220 t class rigid haul truck descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine. Within 1.4 seconds of the light vehicle's beacon appearing over the crest, the operator braked and steered towards the wall. The truck slowed from 37 km/h and struck the right side of LV-118 at 24 km/h. It pushed the light vehicle 18.4 m down the ramp and onto its left side against the inner windrow.

The driver, a survey subcontractor's grade control technician on his third night shift, remained conscious. He was extricated through the rear 27 minutes after the collision. He had a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was flown to the metropolitan trauma centre at 07:40. The light vehicle was destroyed. Its roll-over protection structure deformed but was not breached, and the fastened seatbelt carried loading marks. HT-407, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes, and the main ramp below RL 340 closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.

What the investigation found

The spur was cut under a works order classified as routine earthworks. This excluded it from management of change and from referral to the traffic management plan owner, survey or the contractor's technical services group. It very likely left the intersection without physical controls, a place on the in-cab map or a collision awareness geofence review.

The intersection's sight distance was 41 m against a 96 m stopping distance at the posted speed. The truck very likely could not stop in the distance available. The intersection failed the plan's sight distance rule but had no stop control or warning sign, which likely contributed. Positive calls were required only at the nine listed intersections, and new ones could be listed only at the annual review. This very likely contributed. Neither driver called, consistent with the plan and local practice, which likely contributed.

The light vehicle driver slowed, looked up the ramp, saw nothing and drove out without stopping, although the truck was coming into view. This very likely contributed. His access card was issued without a check that he had completed the pit driving assessment. This likely contributed, although the assessment's content was not available. The S4-LOAD geofence had been extended up the ramp without operators being told. This likely contributed by reinforcing the operator's expectation that a quiet system meant an empty ramp. It did not contribute through a lost alert, because an unsuppressed alert would have sounded only about half a second after the beacon appeared. MSG-14 left the review trigger and the basis for the call list undefined. This likely made inaction the default, although the mine met the guideline and could have acted on its own initiative. Neither driver knew of the other until about six seconds remained.

The 14 night haul roster and the night shift extrication arrangement each increased risk without contributing. Whether dust delayed the sighting, and whether the operator was fatigued, could not be resolved. Brake performance and the road surface were ruled out.

What has been done as a result

At 05:10 on 12/08/2026, the Mining Manager directed a positive call before entering any Stage 4 intersection, listed or not. The spur remains barricaded and closed. On 14/08/2026, survey

measured the other ten unlisted Stage 4 roads, and two below the sight distance minimum were closed pending a stop control. On 28/08/2026, Tarrowen technical services began adding suppressed proximity events to its monthly collision awareness report. The investigation has recommended changes to the plan's listing route, the works order exclusion and the control of geofence changes.

Safety message

A new road becomes an intersection when it first carries traffic, whatever paperwork it was built under. Its defences must be armed when it opens, not at the next annual review. Operators should be told where a detection system is switched off, because a quiet system can teach them to expect an empty road.

The occurrence

Before the night shift

The RL 312 dewatering spur is a 240 m access road in the Stage 4 open pit at Bindarra Gold Mine, cut under works order WO-26-4471 to reach dewatering bore DW-14. The works order classified the job as routine earthworks not requiring management of change. It recorded the earthworks complete on 31/07/2026 and had "N/A" in its handover field. On 02/08/2026 the bore was energised and light vehicles first used the spur. The "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth was removed that day and not replaced.

At the time of the occurrence the intersection had no warning sign on either ramp approach, no give-way or stop control at the spur mouth, no delineation and no posted speed. The fleet management system in-cab map for the Stage 4 pit, last published on 24/07/2026, did not show the spur.

At about 22:10 on 04/08/2026, a dewatering technician drove out of the spur in front of a haul truck descending the main ramp. He reversed and let the truck pass. At 23:48 he submitted hazard report HAZ-2026-0917, rated Medium. He wrote that "you cannot see up the ramp from that road" because of a rise in the ramp and a windrow, noted there was no sign for the trucks and no give way on the spur side, and added: "I do not think they know the road is there." The Dewatering Superintendent closed the report on 06/08/2026 and reduced the rating to Low. He recorded that the crew had been reminded at the 05/08 pre-start "to take care entering the main ramp and to give way to haul trucks at all times". The report was not referred to the mining department, the contractor, the survey department or the traffic management plan owner.

On 09/08/2026 the light vehicle driver, a grade control technician with Hallorn Survey & Geotechnical, was issued a site access card at the Bindarra gatehouse. It was issued against Hallorn's recognition agreement after a 20 minute briefing on muster points, speed limits and personal protective equipment. He had completed Hallorn's two hour online general mining induction on 24/07/2026. He had not completed the Bindarra site-specific supplement or the Bindarra light vehicle pit driving assessment, and the gatehouse did not ask for either. That evening he began the first of three night shifts at Bindarra. He reported that the dewatering foreman drove him to the spur on the first night and told him it was open to light vehicles.

The approach

The haul truck operator was on night 10 of a 14 night swing, driving HT-407, a 220 t class rigid haul truck, between the Stage 4 loading face and the ROM pad. At 02:22:41 on 12/08/2026 dispatch called the Stage 4 crews back to work after crib. On his first cycle the operator made positive calls at the three-sixty crossover and the ROM pad turn-off, two of the nine intersections listed in the traffic management plan. At 02:46:21, returning empty, he called "407, three-sixty crossover, empty, going down through."

The light vehicle driver, in LV-118, had finished a set of levels at the DW-14 collar. He had driven the spur six or seven times over three nights without meeting a haul truck there. He reached the spur mouth at about 02:47. His own estimate of leaving the collar, about ten to three, was later than the recorded collision time. No permit, call-in or board existed by which the shift could know a light vehicle was in the pit, and the night shift supervisor did not know LV-118 was there. The driver's radio was on channel 12 and turned up. He did not recall hearing anything on it, and nobody had told him to make a call before entering the ramp.

HT-407 descended at 39 km/h at RL 384 and 38 km/h at RL 351, with the retarder in the second position. The spur joins the ramp on a descending truck's left, 34 m below a convex grade break that the crews call the crest. The survey found that the operator, at an eye height of 5.80 m, first had line of sight to the spur mouth at 41 m. The top of a light vehicle's 4.5 m beacon whip was in view from 53 m, but not its body. A driver at the spur mouth, at an eye height of 1.45 m, first had line of sight to a descending truck at 38 m. The 1.6 m inner windrow hid the truck's lower body for the last 25 m of the spur.

At 02:47:06.0 the truck was at 37 km/h at RL 317, 53 m above the spur mouth. This was its last steady data point, and the survey places first sight of the beacon there. The operator described a glow on the road and then the whip coming up over the crest, and estimated the light vehicle was about forty metres away. At 02:47:07.4, 38.5 m above the mouth, service brake pressure rose from 0 to 620 kPa in 0.4 seconds. The retarder went to maximum and 14 degrees of right steering was applied. The operator steered towards the wall because the open side fell away steeply, saying: "Right was the only answer." He had known the spur was cut but believed it was still closed, and his in-cab screen showed no intersection at RL 312.

The light vehicle driver reported that he slowed at the mouth, looked right up the ramp and saw nothing. He then turned right to go up the ramp, crossing the descending lane with his right side towards the truck. He did not believe he came to a complete stop, and estimated his speed at about 25 km/h. Neither driver made a radio call before entering the intersection, which was not one of the nine at which the plan required one.

The truck operator recalled some dust hanging on the ramp and could not tell whether it made a difference. The light vehicle driver described the air in the spur as clear.

At 02:47:09.2, with HT-407 at RL 314 and 32 km/h, the collision awareness system logged a proximity event at 21.4 m. Because LV-118 was inside the S4-LOAD suppression geofence, the alert was suppressed and nothing was presented in the cab. The operator had never had an alert anywhere in the pit in fourteen months and did not know the system was switched off there.

The collision

At 02:47:12.0 the front left of HT-407 struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the spur mouth and 3.1 m from the toe of the inner windrow. Without braking, the collision speed would have been 37 km/h. The light vehicle's marks changed from tyre scuff to body panel gouge at 11.2 m, where it went over. It came to rest on its left side against the inner windrow, 18.4 m below the impact point and facing down-ramp. HT-407 stopped 19 m below the intersection at about 02:47:19.

The roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was fastened and its webbing showed loading marks. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two access steps.

The response

At 02:47:41 the operator reported the collision on the pit channel. At 02:48:01 dispatch stopped all Stage 4 units and told him to stay in the cab until traffic was stopped, and he was released at 02:48:44. The emergency response team was paged at 02:48. The operator found the driver conscious and trapped, with the door against the ground. The supervisor told him not to move the driver and to wait for the emergency response team.

The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The paramedic's primary survey and extrication plan were complete at 02:58. The second responder, who must be

present before an extrication, arrived from the accommodation village at 03:07 with the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision. The post-incident review found the extrication was delayed by approximately 9 minutes but that this did not affect the clinical outcome, because the injuries were stable and the patient was conscious throughout. The driver reached the surface medical facility at 03:41 with a fractured left clavicle, three fractured left ribs and a small pneumothorax. He was flown to the metropolitan trauma centre at 07:40.

The Stage 4 pit stood down for 9 hours 15 minutes, and the main ramp below RL 340 was closed for 22 hours. The spur was barricaded. The operator and the shift supervisor tested negative for drugs and alcohol. At 05:10 the Mining Manager directed that a positive call be made before entering any Stage 4 intersection, whether listed in the plan or not.

Context

Personnel information

Haul truck operator

The haul truck operator had 11 years of open pit mining experience, 7 on 220 t class rigid haul trucks. He had been employed by Tarrowen Mining Services for 4 years 2 months and had worked at Bindarra, fly-in fly-out, for 14 months. His verification of competency on the 220 t class unit was issued in 06/2025 and was current. He completed his annual refresher in 02/2026 and held current dozer and water cart verifications.

He completed the site general induction in 06/2025. Its traffic management module, a 23 slide package, named the nine intersections at Appendix B of the traffic management plan but did not address unlisted intersections. He had no traffic management refresher since. No person on site had training on the collision awareness system's suppression geofences.

The Stage 4 haul roster was 14 nights on and 7 off, from 18:00 to 06:30, a 12 hour 30 minute shift with a 30 minute crib. The contractor's fatigue standard permitted up to 14 consecutive night shifts with a minimum 10 hour break, and both limits were met. None of the previous nine shifts extended beyond 12 hours 40 minutes. His voluntary sleep record showed 6 hours 30 minutes of sleep in the day before the occurrence and 5 to 7 hours on each of the preceding nine days. At interview he said he was "Tired, the way you are on night ten. Not falling asleep tired." He added: "I was awake and I was driving properly."

Light vehicle driver

The light vehicle driver was a grade control technician with Hallorn Survey & Geotechnical. He had 11 years in survey, 6 in mining grade control, and 4 years with Hallorn, rotating between four client sites from its regional office. He worked 9 on 5 off, from 18:00 to 06:00, and was on his third night at Bindarra. No duty or rest concerns were identified.

Neither Hallorn's training file nor the gatehouse access card register showed that he had been issued, completed or signed for the Bindarra site-specific supplement, or completed the Bindarra light vehicle pit driving assessment. He could not recall receiving the supplement. [NOT ESTABLISHED: whether the Bindarra site-specific supplement was issued to the light vehicle driver]

Night shift supervisor

The night shift supervisor had 19 years in open pit mining, 8 with Tarrowen Mining Services and 2 years 6 months as supervisor. He was responsible for the haul fleet, excavators, ancillary fleet, dozers and dispatch operator, but not the dewatering crew, the survey contractor or other pit light vehicles. He was also on night 10 of 14.

He knew the dewatering crew used the RL 312 spur, but regarded it as one of many bore tracks rather than an intersection. His pre-start notes for the ten days after it opened did not mention it. He reported that no permit, radio call-in, board or pit entry check told him when a light vehicle was in the pit.

Vehicle information

HT-407

HT-407 was a 220 t class rigid haul truck, commissioned in 03/2021, with 31,480 engine hours. Its last 500 hour service, on 28/07/2026, carried no defects. The brake test on 03/08/2026 gave 0.80

m/s² net, empty, on a 9 per cent descending grade. A repeat on 14/08/2026, before return to service, gave 0.82 m/s² on the same grade. The 11/08/2026 pre-start sheet recorded no defects. No defect had been raised against the brakes, retarder, park brake, lighting, mirrors, cameras or cab glass in the preceding twelve months.

LV-118

LV-118 was a 4WD utility in service since 02/2024, with 61,200 km, last serviced on 17/07/2026. It had a certified roll-over protection structure, a 4.5 m amber beacon whip, reflective marking to the site standard and a UHF radio. The weekly inspection for the week of 09/08/2026 recorded all items serviceable, and the beacon worked when tested after recovery. LV-118 was not dispatched, carried no fleet management system, and had a passive collision awareness system tag.

Fleet management system

The in-cab Stage 4 road network was last published on 24/07/2026 and did not contain the RL 312 spur. Revising it was a survey task raised through a management of change record, and none existed for the spur. The operator said he expected new roads to appear on his screen, as the relocated ROM pad tip head had the night after it moved.

Collision awareness system

The Stage 4 haul fleet system operated at Level 8, operator advisory. Level 9, automated intervention, was available but not enabled at Bindarra. An alert was raised when a tag entered a 60 m ring at a closing rate above 10 km/h. Inside a suppression geofence "no alert is raised at any range, and the system logs a proximity event only once the range falls below 25 m."

The S4-LOAD geofence was drawn on 14/03/2025 around the loading face to stop nuisance alerts near the excavator. On 09/06/2026 the contractor's technical services group extended it northward as a configuration task, carrying the boundary up the main ramp for the first time, to 180 m up-ramp of the RL 312 intersection. The configuration file held only the new boundary, so the size of the extension could not be determined. The change was not recorded in or reviewed against the traffic management plan. Outside a geofence, the system would have alerted when LV-118's tag entered the 60 m ring, at about 02:47:06.5.

In the 12 months to 11/08/2026, HT-407 raised 1,684 alerts, all outside S4-LOAD. Across the fleet, 68 proximity events inside it were suppressed. Nine involved a light vehicle tag on the main ramp below the boundary, all after 09/06/2026. The group's standing report covered alert counts and system health but not suppressed events.

The supervisor learned in about mid 2025 that alerts were suppressed around the dig, but was not aware of the June 2026 extension.

The RL 312 intersection

The main ramp at RL 312 ran approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows. Traffic kept left, so a descending truck ran on the open, pit side. The spur joined on that side, inside a left-hand bend, at an included angle of 71 degrees.

The survey calculated the stopping distance of an empty 220 t class truck at the posted 40 km/h, on a 9 per cent descent with a 1.5 second response, as 96 m. It found: "The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb."

Light, weather and surface conditions

The occurrence was in darkness. The Stage 4 pit had no fixed lighting, and the loading face light plants were 1.1 km away behind the crest. Moonset was at 23:18 the previous evening. The met station recorded a south-east wind of 8 km/h, 9 degrees and no precipitation. The last rain was 26 days earlier, and the investigator found the ramp dry throughout at 06:20.

The water cart last watered RL 280 to RL 400 at 01:10. The standing instruction was a pass every 90 minutes, so a pass was 7 minutes overdue. The RL 328 bench dust monitor, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre from 02:40 to 02:50, against a shift mean of 88 and a site trigger of 250. Nothing at Bindarra measured visibility.

Procedures

Traffic management plan

BIN-TMP-004, owned by the Mining Manager, was issued on 12/11/2025, with review due on 12/11/2026. Clause 4.1 required that "Light vehicles shall give way to heavy vehicles at all times and in all circumstances." Clause 4.3.1 required a positive call before entering any Appendix B intersection. Clause 4.3.2 stated that at other intersections "No call is required."

Clause 6.2 required sight distance at every intersection of not less than the largest vehicle's stopping distance at the posted speed "and in no case less than 120 m". Otherwise a stop control, signalised control or traffic controller was required. Clause 6.4 required a warning sign, give-way control and delineation at every intersection on the Schedule 1 road layout drawing.

Appendix B listed nine intersections and was reissued at the annual review. An intersection was added if it appeared on Schedule 1 and the review assessed it as requiring positive communication. Schedule 1, dated 12/11/2025, did not show the spur. Nothing allowed additions between reviews or required a new road to be referred to the plan's owner. The Stage 4 crib room board carried the 12/11/2025 Appendix B.

Annual reviews were completed on 14/11/2023, 10/11/2024 and 12/11/2025. No out-of-cycle review had been held under clause 9.1 in the plan's eleven year history. Since 12/11/2025, 14 new roads had been cut in Stage 4. The 3 cut under a management of change record were referred to the owner, and none of the 11 cut under a works order was.

Management of change

ORR-PRO-022 clause 5.2 required a management of change record before "a change to the mine's road network". Clause 5.3 excluded "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced." Under clause 5.4 the change owner's impact assessment included referral to the plan owner, to survey for the fleet management system road network, and to technical services for the collision awareness system configuration. The works order form had no equivalent fields.

Contractor induction

ORR-STD-009 clause 7.3 allowed a contractor's general induction to be accepted where the contractor held a recognition agreement, as Hallorn had since 2023. Clause 7.4 excluded the site-specific supplement and pit driving assessment, which "shall not be substituted". Clause 7.5 made the engaging department responsible for verifying completion before card issue. The gatehouse issued cards against the recognition agreement alone, and the card system did not distinguish who could drive in an operating pit.

Regulatory guidance

Mines Safety Guideline MSG-14, revision 4, was issued by the State Mines Inspectorate under regulation 6.22 of the Mining Operations Safety Regulations 2018. The regulation required only that "the operator shall manage the risks arising from the interaction of vehicles at the mine".

Clause 3.2 required review within twelve months and after "any significant change", which it did not define. Clause 5.1 required the operator to nominate intersections needing positive communication, with no basis or minimum. Under clause 3.5 the plan is a site document, not approved by the Inspectorate. Clause 5.4 allowed light vehicle detectability by any of five means. On the day, Bindarra's plan met the guideline on its face.

Emergency response arrangements

The night shift had one mines rescue paramedic, based at the surface medical facility. A second responder was required before extrication and was called from the accommodation village, 11 km from the gatehouse, in the second response vehicle. That vehicle was garaged at the village and carried the hydraulic rescue tool.

Related occurrences

Bindarra hazard reports for the 24 months to 11/08/2026 included four relevant reports, among them HAZ-2026-0917. The other three, from 03/2025, 09/2025 and 02/2026, each described a light vehicle and haul truck meeting unexpectedly at an unlisted intersection, at three different locations. All four were closed within three days with a crew reminder. None was referred to the plan's owner or led to a change to Appendix B, Schedule 1 or any physical control.

Safety analysis

External Influences

Guideline sets no review trigger and no basis for the positive-call list

Bindarra's traffic management plan was made under Mines Safety Guideline MSG-14, revision 4, issued by the State Mines Inspectorate in 2021 under regulation 6.22 of the Mining Operations Safety Regulations 2018. The regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". The guideline is one way of meeting that requirement. Three of its clauses bear on this occurrence:

- "The plan shall be reviewed at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern." [MSG-14, clause 3.2]
- "The operator shall nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry, and shall record them in the plan." [MSG-14, clause 5.1]
- "The plan is a site document. It is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection." [MSG-14, clause 3.5]

The guideline does not define a significant change and gives no example of one. It sets no minimum number of positive-call intersections and no basis for choosing them. Because the plan is not reviewed outside the mine, both judgements rest with the operator alone. These features of the guideline were virtually certain to have existed, because they are documentary and not in dispute.

On the day of the occurrence Bindarra's plan met the guideline on its face. It had been reviewed on 12/11/2025, inside twelve months, and its Appendix B nominated nine intersections. The mine was not in breach of MSG-14. The Inspectorate did not fail to do anything the guideline gave it to do. The matter considered here is the breadth of the discretion the guideline leaves to the operator. Clause 5.4, which allows a light vehicle to be made detectable by any one of five means, is part of that context but is not part of this factor.

The guideline bears on the occurrence through one path, which is how an intersection reaches the positive-call list. The plan reproduced the guideline's structure without adding to it. Clause 9.1 repeated the requirement to review "annually, and following any significant change to the mine's road network", again without a definition. The note to Appendix B provided that the appendix "is reviewed and reissued at the annual review of this plan". It also provided that an intersection is added only where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication. Nothing in the plan allowed an intersection to be added between annual reviews. Nothing required a new road to be referred to the plan's owner.

The plan's review history shows how that worked in practice. The plan had never had an out-of-cycle review in its eleven years. Fourteen new roads had been cut in the Stage 4 pit since the November 2025 review. The three cut under a management of change record were referred to the plan's owner, and the eleven cut under a works order were not. Four hazard reports in the 24 months before the occurrence described a light vehicle and a haul truck meeting unexpectedly at an intersection not listed in Appendix B. The four reports came from four different locations, one of them the RL 312 spur. Each was closed within three days with a reminder to the crew. None reached the plan's owner, and none led to a change to Appendix B, Schedule 1 or any physical control.

The argument for influence is that a defined trigger would have required the plan to be reopened when a road was added to a haul road network. A stated basis for nomination would also have given an intersection that failed the plan's own sight distance rule a claim on the positive-call list. As written, MSG-14 left both terms to a site document reviewed once a year, with no one outside the mine examining either.

The argument against influence is also substantial. The guideline did not prevent Bindarra from acting. Clause 9.1 of the plan allowed an out-of-cycle review at any time, and MSG-14 does not stop a mine from listing every intersection it has. On this reading the mine did not fail to act because the guideline was vague. It failed to act because nothing inside the mine put the question to the plan's owner, and the guideline is background only.

Neither argument displaces the other. The mine could have defined its own trigger and its own basis for nomination. However, by leaving both undefined and unexamined outside the mine, the guideline made inaction the default outcome when a road was cut between reviews. On balance it was likely that the absence of a review trigger and of a basis for the positive-call list contributed to the arrangement under which positive calls were required only at listed intersections, added only at the annual review. That arrangement was found very likely to have contributed to the occurrence, and it is analysed under Risk Controls. The finding for the guideline is held at likely and no higher, because the evidence against it is substantial and the balance in its favour is narrow.

The guideline has no bearing on why the spur was opened under a works order without referral. Whether a works order is exempt from management of change was set by the mine's own procedure, ORR-PRO-022 clause 5.3, and not by MSG-14. That matter is analysed under Organisational Influences.

Organisational Influences

Spur opened under a works order, so nothing was referred to anyone

The mine's management of change procedure, ORR-PRO-022, requires a record before "a change to the mine's road network" (clause 5.2). Clause 5.3 then excludes "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced." Works order WO-26-4471 cut 240 m of road to bore DW-14 at RL 312 and classified the work under that exclusion. It recorded management of change as not required, and its handover field read "N/A". The procedure, the works order and the road register agree, and it is virtually certain that the spur was opened without a management of change record.

That classification mattered because the procedure's referrals sit only inside the management of change record. Clause 5.4 sends a change to three parties:

- the traffic management plan owner
- the survey department, for the fleet management system road network
- the contractor's technical services group, for the collision awareness system configuration.

The works order form has no equivalent fields. None of the three referrals was made. The in-cab road network published on 24/07/2026 was still current on the night and showed no intersection at RL 312. The S4-LOAD suppression geofence had been carried up the ramp on 09/06/2026 and already covered the place where the spur joined it.

A management of change record would very likely have changed the conditions at the intersection. The plan's owner would have received a new intersection that failed the clause 6.2 sight distance rule, and so needed a stop control and the clause 6.4 signs. The technical services group configures the geofence that the intersection had been cut inside. Both referrals are standing parts of the

procedure. The three Stage 4 roads raised under a management of change record since the 12/11/2025 review were all referred to the plan's owner, and none of the eleven raised under a works order was. A referral is not itself a control, however. The plan's owner might have left the spur to the annual review, and no out-of-cycle review has been held in the plan's eleven years.

This factor led to two risk control findings: the absence of a stop control or warning sign at the intersection, and the suppression of collision alerts at the spur. Both are argued under Risk Controls. It did not lead to the positive-call finding. A referral could not have changed the rule that limits calls to listed intersections, and the owner's power to direct a call between reviews had never been used.

Access cards issued with no check of the pit driving assessment

ORR-STD-009 allows a contractor's general induction to stand in for the Bindarra induction where the contractor holds a recognition agreement (clause 7.3). Hallorn Survey & Geotechnical has held one since 2023. Clause 7.4 states that this equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment". Clause 7.5 makes the engaging department responsible for verifying both before a card is issued.

The standard sets the right requirement. The gatehouse, however, issues cards against the recognition agreement alone. It holds no record of either item and is not asked for one. Nothing in the access card system distinguishes a person who may drive in an operating pit from one who may not.

The light vehicle driver's card was issued on 09/08/2026 after a 20 minute briefing on muster points, speed limits and personal protective equipment. His training file records the Hallorn online module of 24/07/2026 and no pit driving assessment. He had never heard of the assessment and had never seen BIN-TMP-004. There is no record of the site-specific supplement being issued, completed or signed for, and whether he received it was not established. It is virtually certain that the card was issued with no check of the assessment.

The evidence on influence pointed both ways. The assessment is where a light vehicle driver learns how the pit's traffic rules work in practice, including how to approach an intersection with no view up the road. Against that, the driver was already applying the assessment's first rule, that light vehicles give way to heavy vehicles at all times. He slowed and looked before he went. The plan requires no call at an unlisted intersection, so the assessment would not have taught him to call at RL 312. His statement that he would have called if told to does not reach that far.

The assessment's contribution lies elsewhere. It teaches a driver to stop at a mouth he cannot see out of and look again, rather than to slow. On the recorded times, a stop at the mouth would very likely have shown him the truck. The content of the assessment was not available, and the driver was already applying its first rule. For those reasons, the absence of the check is put no higher than likely to have influenced the light vehicle's exit from the spur without stopping.

Haul operators rostered for fourteen consecutive 12.5 hour night shifts

The contractor's Stage 4 haul roster is 14 nights on and 7 off, 18:00 to 06:30 local, with a 30 minute crib. It sits inside a fatigue management standard that allows up to 14 consecutive night shifts with a minimum 10 hour break. The haul truck operator was on night 10 of 14. The collision at 02:47 was 8 hours 47 minutes into the shift, inside the window of lowest circadian alertness. Every limit in the standard was met, and no previous shift ran beyond 12 hours 40 minutes. The night shift supervisor works the same pattern. The roster is documentary, and its existence is virtually certain.

A roster of this kind degrades detection and response first, and those are the tasks needed to pick out a hazard in an unlit pit. The recorded data does not show that effect here. The operator braked

1.4 seconds after the beacon came into view, inside the 1.5 seconds assumed in the plan's own stopping distance calculation, and his braking and steering were correct. At the recorded 0.78 m/s² from 53 m, a response time of zero would still have ended in a collision, at about 17 km/h instead of 24 km/h. The download shows no decision or omission for the roster to have acted through. It is therefore unlikely that the roster influenced the occurrence. The operator's own state on the night is a separate matter and is discussed under Local Conditions.

The roster is still likely to have increased risk. It is a standing exposure built into every swing, not a feature of one night. It puts each operator at the wheel of a 220 tonne vehicle in the small hours of a tenth, twelfth and fourteenth consecutive night. On those nights, the defence against an unlit hazard at an unlisted intersection is the operator's own detection and response. At an intersection with a longer sight line than RL 312, a slower response could decide whether a truck stops in time.

Risk Controls

Collision alerts suppressed at the spur by a geofence operators knew nothing of

The collision awareness system on HT-407 operated at Level 8, operator advisory. Level 9 intervention was available on the fleet but not enabled at Bindarra. The system raised an alert when a tag entered the 60 m ring at a closing rate above 10 km/h, except inside a suppression geofence. Inside a geofence it raised no alert at any range.

The S4-LOAD geofence was drawn around the Stage 4 loading face on 14/03/2025 to stop nuisance alerts. On 09/06/2026 the contractor's technical services group extended it northward as a configuration task. That extension carried the boundary up the main ramp for the first time, to a point 180 m up-ramp of where the RL 312 intersection was later cut. The change was not recorded in the traffic management plan and was not reviewed against it. The event log recorded the light vehicle at 21.4 m at 02:47:09.2, with the alert suppressed.

No training record on site, for any person, covered the geofences. The truck operator and the night shift supervisor each said they did not know the system was off on the ramp. It is extremely likely that the alerts were suppressed at the spur without operators knowing. The remaining doubt concerns what operators knew, because that part rests on two accounts and on the absence of any training record or notice.

The alert itself would have made no difference. Unsuppressed, it would have sounded at about 02:47:06.5. That was half a second after the beacon came into view and about a second before the brakes were applied. Any influence therefore lies in the operator's expectation. HT-407 raised 1,684 alerts in the preceding twelve months, none of them inside the geofence. For the ten months before the extension the ramp was quiet because nothing was there. After June it was silent regardless, and nine suppressed events involved a light vehicle on that stretch of ramp. In his own report, before he was asked, the operator wrote that he had taken a quiet system to mean an empty pit.

Against this, he was descending at the speed every operator used on that ramp. The remedy he named was a sign, not the system. On balance, it is likely that the suppression contributed by reinforcing the operator's expectation that the ramp was empty. That effect is real but could not be measured. The suppression arose from the spur being opened under a works order, which is discussed under Organisational Influences.

Positive calls required only at listed intersections, added only at the annual review

Under BIN-TMP-004, clause 4.3.1 required a positive call before entering an intersection listed at Appendix B. Clause 4.3.2 required no call anywhere else. An intersection reached Appendix B only by appearing on the Schedule 1 road layout drawing, and that drawing was reissued only at the

annual review. No provision allowed an intersection to be added between reviews. No provision required a new road to be referred to the plan's owner.

The RL 312 spur appeared on neither document. Eleven of the fourteen roads cut in Stage 4 since the 12/11/2025 review were never referred. No out-of-cycle review had been held in the plan's eleven years.

The pit channel recording shows the practice that resulted. HT-407 called at the RL 360 crossover in both directions and at the ROM pad turn-off. Its last call was at 02:46:21, 51 seconds before the collision, and it made no call at the spur. The operator, the light vehicle driver and the supervisor each described the same practice: calls were made at the nine intersections on the crib room board and nowhere else.

It is virtually certain that this limitation existed. It is very likely that it contributed. The operator called at every listed intersection he passed that night, and the light vehicle was on the open channel. The remaining doubt is that a call made but not acknowledged would still have left a driver deciding whether to proceed. The earliest date on which the spur could have been listed was 12/11/2026. The guideline behind this arrangement is discussed under External Influences. The absence of a call by either driver is discussed under Individual Actions.

No stop control or warning sign at an intersection failing the sight distance rule

Clause 6.2 applied to every intersection, listed or not. The survey gave a sight distance of 41 m against a stopping distance of 96 m, so the clause required a stop control, signalised control or traffic controller. Clause 6.4 required signs, a give-way control and delineation, but only at intersections shown on the road layout drawing. At the time of the occurrence the intersection had no warning sign, no stop or give-way control, no delineation and no posted speed. The construction sign was removed on 02/08/2026, and nothing replaced it. HAZ-2026-0917 had named the missing sign and the missing give-way. It is virtually certain that these controls were absent.

The truck was within the light vehicle driver's 38 m of view from about 02:47:07.4. That was nearly two seconds before he reached the mouth. A full stop there would very likely have shown him the truck. The technician on 04/08 reversed into the spur and let a truck pass in similar circumstances. The operator said that a sign on his side of the crest would have had him at twenty and looking for the light vehicle.

Against this, 38 m is about four seconds of a truck at 37 km/h, which is a thin margin even from a standstill. The driver described already slowing and looking. The effect of a sign rests on the operator's view after the event. Because the in-cab map did not show the intersection, a sign would also have been the only cue. It is likely that the absence of these controls contributed. The intersection geometry is discussed under Local Conditions.

Night shift extrication waits on a second responder called from the village

The night shift was rostered with one mines rescue paramedic. The second responder, who must be present before an extrication begins, was called in from the village 11 km away in the vehicle that carried the hydraulic rescue tool. The post-incident review measured a delay of approximately 9 minutes, from 02:58 to 03:07. It found that the delay did not affect the clinical outcome. The collision and the injuries were complete before the team was paged, so it is unlikely that the arrangement affected this occurrence.

It is still very likely to matter. The driver was conscious, stable, restrained and protected by an intact roll-over structure. A fire, an unconscious casualty or an airway problem would turn the same wait

into a different outcome. On night shift the extrication capability was one person plus a drive from the village, and nothing in the arrangement changed with the severity of the event.

Local Conditions

Sight line of 41 m at the spur against a 96 m stopping distance

The survey of the intersection established the geometry with certainty. The spur mouth sat 34 m below the crest. From a truck operator's eye height of 5.80 m, the spur mouth first came into view at 41 m. The top of the light vehicle's 4.5 m beacon whip came into view at 53 m, but the vehicle's body did not. From the spur mouth, the light vehicle driver first had line of sight to a descending truck at 38 m. The 1.6 m inner windrow hid the truck's lower body for the last 25 m of the spur's approach.

Against this, a 220 t class truck travelling empty at the posted 40 km/h on that grade needed 96 m to stop. That is more than twice the distance in which the spur mouth could be seen. At 41 m, a truck at the posted speed could not stop for anything at this intersection, however alert its operator.

The occurrence bore this out. HT-407 was travelling at 37 km/h, and on brakes alone at the recorded 0.78 m/s² it needed about 68 m. With twice the sight line, the same 1.4 second response and the same braking would have stopped the truck short of the spur. The dewatering technician had described the same view seven days earlier: "you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything."

It was virtually certain that the available sight distance at the intersection was 41 m against a stopping distance of 96 m. This very likely contributed to neither driver being aware of the other until the truck came over the crest. The geometry produced a collision only once a vehicle was in the intersection. The defences that should have kept the two vehicles apart are argued under Risk Controls.

Airborne dust on the ramp above the intersection during the truck's approach

Several things pointed to dust above the crest:

- The ramp's last water cart pass was 1 hour 37 minutes before the occurrence. The standing interval was 90 minutes, so the next pass was 7 minutes overdue.
- The fixed dust monitor on the RL 328 bench sits 180 m up-ramp of the intersection and above the grade break. In the 02:40 to 02:50 interval it logged PM₁₀ at 142 micrograms per cubic metre, against a shift mean of 88.
- The truck operator described dust hanging on the descent.
- The truck was descending behind its own lights into an unlit pit.

Two things pointed the other way. The reading was well below the site trigger of 250. It measured particulate at the monitor head, not what could be seen from either cab, and nothing at Bindarra measures visibility. The light vehicle driver was also firm that the air was clear. However, his account described the spur below the windrow, where he was moving slowly or stopped. It did not describe the ramp above the crest where the truck was. On balance, airborne dust was likely present on the ramp above the intersection.

Whether the dust delayed the operator's sighting of the beacon could not be determined.

- **Argument that it did:** dust hanging in a truck's lights over a crest is the condition in which a driver picks out a beacon late.

- **Argument that it did not:** the download places the truck at 53 m at its last steady data point, before the brake application. That is the first point at which the survey says the beacon could be seen at all, even in clear air. A sighting at the geometric limit leaves the dust nothing to explain. The operator himself would not say that the dust made a difference. His 1.4 second response from that point was inside the 1.5 seconds allowed in the plan's stopping distance.

The case against an effect is strong. It does not exclude one, because the moment of sighting was inferred from the data point and the survey, not recorded. A delay of a fraction of a second at the edge of the truck's lights therefore cannot be ruled out. No measurement exists of what could be seen from either cab on the night, so the question cannot be answered with the available evidence. It was about as likely as not that the dust affected the sighting. The matter remains unresolved and is not carried into the findings.

Ramp surface wet or slippery at and above the intersection

It was very unlikely that the ramp was wet or slippery. No rain fell that night or in the previous 26 days. The investigator found the surface dry throughout at 06:20. The braked tyre scuff ran continuously for 58 m with no sign of a locked or sliding wheel. The truck achieved 0.78 m/s², close to its tested dry figure of 0.80 m/s².

Haul truck operator fatigued during the approach to the intersection

Several things suggested fatigue. The operator was on night 10 of a 14 night swing and 8 hours 47 minutes into a 12 hour 30 minute shift. The collision was at 02:47, in the window of lowest circadian alertness. His sleep record, which he kept voluntarily, showed between 5 and 7 hours on each of the ten days before, and 6 hours 30 minutes on the last. He described himself as tired in the ordinary way of a tenth night. The task on the approach was detection in an unlit pit, which is the task fatigue degrades first.

Other evidence argued against impairment. The operator said he was not sleepy and was driving properly. Every limit in the contractor's fatigue standard was met. His recorded performance showed none of the usual signs. He responded in 1.4 seconds, and his braking and steering were correct and immediate.

On this evidence, it was about as likely as not that the operator was fatigued to a degree that affected his performance. A detailed fatigue assessment would be needed to settle the question, and none was carried out. Even if the operator had been fatigued, it would not have changed the outcome. Braking at 0.78 m/s² from 53 m with no response delay at all would still have ended in a collision, at about 17 km/h instead of 24 km/h. The matter remains unresolved and is not carried into the findings. The risk posed by the roster itself is considered under Organisational Influences.

Individual Actions

Neither driver made a positive call before entering the intersection

It was virtually certain that neither driver made a positive call before entering the RL 312 intersection. The pit channel was logged continuously from the return from crib to two minutes after the collision, and the whole period was reviewed. There was no transmission by either vehicle about the spur, the dewatering bore, or a light vehicle being in the Stage 4 pit. After logging on at 18:04 at the start of shift, LV-118 made no further transmission that shift. Interviews with the truck operator, the light vehicle driver and the night shift supervisor confirmed this.

The truck operator called as the plan required. HT-407 called at each listed intersection it passed, and each call was acknowledged. Its last call was "407, three-sixty crossover, empty, going down

through" at 02:46:21, 51 seconds before the collision. BIN-TMP-004 clause 4.3.2 states that at intersections not listed at Appendix B "No call is required", and the RL 312 intersection was not listed.

The three people interviewed each described the same practice independently. Calls were made at the nine intersections on the crib room board and nowhere else. The supervisor stated that he had never corrected anyone for not calling elsewhere. He also said that requiring calls everywhere would make the channel unusable. Neither driver departed from the plan or from the practice the pit followed every shift. The limits of the positive-call requirement are discussed under Risk Controls and External Influences.

It was likely that the absence of a call contributed to neither driver being aware of the other until the truck came over the crest. The other vehicle, dispatch and the supervisor would all have heard a call from either vehicle. The dispatcher was already tracking HT-407 closely enough to tell the excavator operator it was "about four minutes off you". The contribution falls short of certainty because a call gives knowledge, not separation. Even with each driver aware of the other, the two vehicles would still have had to settle which went first across a 41 m sight line.

Light vehicle drove out of the spur onto the ramp without stopping

It was virtually certain that the light vehicle driver drove out of the spur onto the ramp without stopping. He stated that he slowed at the mouth, looked right up the ramp, saw nothing and drove out, turning right to go up the ramp. He estimated his speed at about 25 km/h and did not believe he had come to a complete stop. When his account was read back to him, he corrected it from stopped to slowed, which was against his own interest. His account agreed with the collision awareness log and with the scene. The log placed LV-118 at the spur mouth at 02:47:09.2 with the truck 21.4 m above it. At the scene, the impact point was 6.2 m into the ramp, and the impact occurred 2.8 seconds after the log entry.

The timing explains why his look found nothing. From the mouth, his eye height gave 38 m of sight line up the ramp. Before he reached the mouth, the inner windrow hid the truck's lower body for the last 25 m of the spur. On the recorded times, his look came a second or two before the truck came over the crest, when the truck was beyond anything his position allowed him to see. HT-407 was within 38 m of the mouth from about 02:47:07.4, nearly two seconds before LV-118 reached it.

The driver described himself as applying clause 4.1, which requires light vehicles to give way to heavy vehicles at all times. Nothing at the mouth indicated that the intersection needed more than that. There was no stop or give-way control, no stop line, no warning sign and no posted speed. He had not completed the Bindarra pit driving assessment and had not seen the traffic management plan. These conditions are discussed under Risk Controls and Organisational Influences.

Driving out without stopping placed the light vehicle across the descending truck's path. It was extremely likely that this action contributed to the collision. Had he stopped at the mouth and looked again, the truck was already within his view. He would very likely have seen it and let it pass. The dewatering technician had done the same at this intersection on 04/08/2026 by reversing back into the spur. The contribution falls short of certainty because a stop is only as useful as the view from it, and that view was 38 m.

Truck operator braked and steered right within 1.4 seconds of the beacon appearing

It was virtually certain that the truck operator braked and steered right within 1.4 seconds of the beacon appearing. The fleet management system download recorded the last steady data point at

02:47:06.0, at 37 km/h and 53 m above the spur mouth. That is where the survey placed the first sight of a light vehicle's beacon over the crest. From 02:47:07.4, at 38.5 m above the mouth, three inputs began together. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder went to maximum, and 14 degrees of right steering was applied. The response of 1.4 seconds was within the 1.5 seconds allowed in the plan's stopping distance calculation. Mean deceleration to the collision was 0.78 m/s² net of grade, against the unit's tested 0.80 m/s². The braked scuff at the scene deviated 1.4 m to the right over its first 20 m, which corroborated the download.

It was very likely that this response reduced the severity of the outcome. Impact occurred at 24 km/h, six seconds after the beacon came into view. Without braking, the impact speed would have been 37 km/h. Impact energy varies with the square of speed, so the braked impact carried about 0.42 of the energy of an unbraked one. The steering turned the truck away from the open side of the road, where the pit edge dropped more than 100 m. The push therefore ran down the ramp, and the light vehicle came to rest against the inner windrow rather than being driven through it.

The truck stopped 19 m below the intersection. The operator then reported the collision and stayed in the cab, as dispatch instructed, until the traffic was stopped. He then stayed with the driver until the response team arrived.

Occurrence Events

Neither driver aware of the other until the truck came over the crest

It was extremely likely that neither driver knew the other was there until the truck came over the crest. It was also extremely likely that this lack of awareness led to the collision. Each condition examined in the earlier subsections arrives at this point, and it is best read as the single mechanism through which those conditions acted.

From the cab of HT-407, nothing indicated an intersection at RL 312. The truck was descending empty at 37 km/h. Its in-cab road network had last been published on 24/07/2026, a week before the spur was cut, and showed the main ramp with no intersection there. There was no warning sign on the approach and no call on the channel. The collision awareness system raised no alert, although it later logged LV-118 at 21.4 m.

The light vehicle driver was approaching a spur mouth from which, on the survey, a descending truck first came into view at 38 m. The inner windrow, 1.6 m high, took the truck's lower body out of his view before that. He had no give-way control, made no call and had no knowledge of the truck. His look up the ramp came before the truck was within his view. The night shift supervisor did not know a light vehicle was in the pit. There was no permit, call-in or board by which he could have known.

The first sighting available to the truck operator was the light vehicle's beacon over the crest at 53 m. The survey placed the top of the 4.5 m beacon whip in view from that distance, with the vehicle's body still hidden. The truck's last steady data point, at 02:47:06, puts it at that position six seconds before the impact. The radio log, the download, the collision awareness system log and all three interviews contain no indication that either driver knew of the other before then. The one residual uncertainty is that the light vehicle's movements before it reached the spur mouth rest on the driver's own account alone.

Once the two vehicles had converged with six seconds remaining, nothing between that moment and the impact could have prevented the collision. The truck operator's response in those six seconds is discussed under Individual Actions. The suppressed alert and the missing intersection controls are discussed under Risk Controls, and the sight line under Local Conditions. Airborne dust on the approach could not be resolved and is discussed under Local Conditions.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

It was virtually certain that the front left of HT-407 struck the right side of LV-118 at 24 km/h. It was virtually certain that this impact was the means by which the convergence became an injury. The download, the damage pattern and the debris field agree.

The impact occurred at 02:47:12, 4.6 seconds after the brakes were applied. It was 6.2 m into the ramp from the spur mouth and 3.1 m out from the toe of the inner windrow. The truck's accelerometer logged 1.9 g longitudinal. Damage to LV-118 was concentrated on the right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with none above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The intersection geometry made a side impact the likely form of any collision there. The spur joins the ramp on a descending truck's left, the open side, on the inside of a left-hand bend at an included angle of 71 degrees. A vehicle turning right out of the spur to go up the ramp must cross the descending lane with its right side towards a truck coming down. That was the position LV-118 was in. The truck operator's braking reduced the impact speed from the 37 km/h it would otherwise have been to 24 km/h.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

It was virtually certain that LV-118 was pushed about 18 m down the ramp and rolled onto its left side. It was virtually certain that the roll-over was how the collision produced the injuries. Scene measurements, the vehicle examination and the medical evidence agree.

A continuous gouge ran 18.4 m down-ramp from the impact position. At 11.2 m it changed from tyre scuff to body panel gouge, which is where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He was conscious throughout. The door on the resting side was against the ground, so he was extricated through the rear of the vehicle.

Two protective features worked. The driver's seatbelt was found fastened, with loading marks on the webbing. The roll-over protection structure was deformed inward at the right rear mounting but was not breached. The injuries were those a side impact and roll-over produce when the structure holds and the restraint is worn. The time taken to extricate the driver is discussed under Risk Controls.

Service brakes delivered less than their tested deceleration during the stop

It was very unlikely that the service brakes delivered less than their tested deceleration, because the 0.78 m/s² achieved during the occurrence lay within the scatter of the 0.80 and 0.82 m/s² measured in tests on 03/08/2026 and 14/08/2026, and no brake, retarder or park brake defect had been raised against the unit in the preceding twelve months.

Findings

From the evidence available, the following findings are made with respect to the collision between a haul truck and a light vehicle on the main ramp of the Stage 4 open pit at Bindarra Gold Mine on 12/08/2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

A finding stated without a likelihood term was established as fact.

Contributing factors

- It is virtually certain that the RL 312 dewatering spur was opened under works order WO-26-4471 as routine earthworks excluded from management of change, so that none of the referrals in ORR-PRO-022 clause 5.4 was made to the traffic management plan owner, the survey department or the contractor's technical services group, and very likely that this contributed by leaving the new intersection without the controls, the in-cab map update and the geofence review that a referral would have prompted. (Safety Issue)
- It is very likely that collision alerts at the spur were suppressed by the S4-LOAD geofence, which had been extended up the main ramp on 09/06/2026 without operators being told. It is likely that this contributed by reinforcing the haul truck operator's expectation that a quiet system meant an empty ramp, although an unsuppressed advisory alert would have sounded only about half a second after the beacon came into view. (Safety Issue)
- Surveyed sight distance at the RL 312 intersection was 41 m against a stopping distance of 96 m for an empty 220 t class truck at the posted 40 km/h, a condition that was virtually certain and that very likely contributed, because the truck could not stop for a light vehicle in the distance available.
- MSG-14 neither defined the significant change that requires a traffic management plan to be reviewed nor set a basis for nominating positive-call intersections, which is virtually certain. On the balance of the evidence rather than as a settled conclusion, this likely contributed by making no listing between annual reviews the default, although Bindarra's plan met the guideline on its face and the mine could have held an out-of-cycle review or listed every intersection on its own initiative. (Safety Issue)
- Under BIN-TMP-004, positive calls were required only at the nine intersections listed at Appendix B, and an intersection could be added only at the annual review, which is virtually certain. This very likely contributed because the RL 312 intersection was unlisted and could not have been listed before 12/11/2026. (Safety Issue)
- It is virtually certain that neither driver made a positive call before entering the RL 312 intersection, where the plan required none and local practice matched the plan. It is likely that this contributed, because a call would have given each driver knowledge of the other, though not separation over 41 m of sight line.
- The light vehicle driver's access card was issued against the contractor's recognition agreement with no check that he had completed the Bindarra light vehicle pit driving assessment, because the access card system had no field for it, which is virtually certain. This likely contributed by leaving him without the assessment's instruction on approaching an intersection he could not see out of, although no copy of the assessment's content was available. (Safety Issue)
- It is virtually certain that the RL 312 intersection had no stop control, give-way control, delineation or warning sign, even though its sight distance failed BIN-TMP-004 clause 6.2. It is likely that this contributed, because a stop control would have held the light vehicle

at the mouth while the truck came into view, and a warning sign would have told the operator there was an intersection below the crest. (Safety Issue)

- The light vehicle driver slowed at the spur mouth, looked up the ramp, saw nothing and drove onto the ramp without coming to a complete stop, which is virtually certain. This very likely contributed, because the truck was within his 38 m of view from about 02:47:07.4, nearly two seconds before he reached the mouth.
- Neither driver was aware of the other until the truck came over the crest, about six seconds before the impact, which is very likely, and this very likely contributed to the collision.
- At 02:47:12.0 the front left of HT-407 struck the right side of LV-118 at 24 km/h, which is virtually certain, and it is virtually certain that this contributed to the driver's injuries.
- LV-118 was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow, which is virtually certain. It is virtually certain that this contributed to the driver's injuries.

Other factors that increased risk

- The contractor's Stage 4 haul roster put operators at the wheel for fourteen consecutive 12.5 hour night shifts, which is virtually certain. Although this was not found to have contributed to this occurrence, it likely increased risk by placing operators in the small hours of a tenth, twelfth and fourteenth consecutive night, where detection of an unlit hazard at an unlisted intersection depends on their own response. (Safety Issue)
- It is virtually certain that extrication on night shift could not begin until a second responder arrived from the accommodation village, 11 km away, with the hydraulic rescue tool. It is very likely that this increased risk, because the arrangement does not change with the severity of the casualty, although the post-incident review found that the delay of approximately 9 minutes on this occasion did not affect the clinical outcome. (Safety Issue)

Other findings

- The ramp surface at and above the intersection was examined and a wet or slippery surface was ruled out. No precipitation was recorded, the last rain was 26 days earlier, the ramp was found dry at 06:20, and the truck's recorded deceleration of 0.78 m/s² net matched its dry-grade tests.
- Service brakes delivering less than their tested deceleration during the stop was examined and ruled out. The recorded 0.78 m/s² net fell within the scatter of brake tests of 0.80 m/s² on 03/08/2026 and 0.82 m/s² on 14/08/2026, and no brake or retarder defect had been raised in the preceding twelve months.
- The haul truck operator braked and steered right within 1.4 seconds of the light vehicle's beacon coming into view, inside the 1.5 seconds allowed in the plan's stopping distance calculation. This reduced the impact speed from 37 km/h to 24 km/h and turned the truck towards the wall, away from the pit edge.

Safety issues and actions

Guideline sets no review trigger and no basis for the positive-call list

Mines Safety Guideline MSG-14 required a traffic management plan to be reviewed after any significant change to the road network but did not define that term. It also required the operator to nominate the intersections at which positive communication was required, without setting a minimum or any basis for the nomination. The plan was a site document. It was not approved by the Inspectorate and was not submitted to it except on request after an occurrence. Bindarra's plan met the guideline on its face, and the mine was not in breach.

The safety significance lies in the breadth of the discretion, not in any act or omission by the Inspectorate. Both the review trigger and the basis for listing were left to the same site document, which was reviewed once a year, and nobody outside the mine examined either. As a result, a new intersection cut between annual reviews had no defined route onto the call list. At Bindarra, eleven of the fourteen new Stage 4 roads cut since the last review had never been referred to the plan's owner. The guideline likely contributed to the arrangement under which positive calls were required only at listed intersections, added only at the annual review. That finding is held on the balance of the evidence rather than as a settled conclusion, because the mine could have held an out-of-cycle review or listed every intersection on its own initiative.

No safety action has been recorded in response to this safety issue.

Spur opened under a works order, so nothing was referred to anyone

The RL 312 dewatering spur was cut under works order WO-26-4471, which classified the work as routine earthworks within the approved Stage 4 design. Under ORR-PRO-022 clause 5.3, that classification removed the requirement for a management of change record. It is virtually certain that, as a result, none of the three referrals at clause 5.4 was made. The traffic management plan owner, the survey department and the contractor's technical services group were not told of the new intersection.

Everything that should follow a new road into an operating pit sits inside the management of change record. Without it, the intersection had no stop control or warning signs. The in-cab road network published on 24/07/2026 showed no intersection at RL 312. Nobody reviewed the S4-LOAD suppression geofence, which already covered the junction. It is very likely that this contributed to the collision. The problem was not confined to one road. Since the 12/11/2025 review, fourteen new roads have been cut in Stage 4. The three raised under a management of change record were referred to the plan's owner, and none of the eleven raised under a works order was.

Proposed safety action

On 01/09/2026 the investigator in charge recommended one of two changes. Either ORR-PRO-022 clause 5.3 should stop excluding earthworks that create a new intersection with a haul road, or the works order form at Schedule 4 should carry the three referrals held at clause 5.4. [NOT ESTABLISHED: organisation to which the recommendation was directed]. A response was due by 30/09/2026, and the recommendation was open. [NOT ESTABLISHED: response received, or action taken in reply]

Access cards issued with no check of the pit driving assessment

The Bindarra gatehouse issued access cards to contractor personnel against their employer's recognition agreement alone. It did not check whether a person who would drive in an operating pit had completed the Bindarra light vehicle pit driving assessment.

ORR-STD-009 clause 7.4 states that equivalence under a recognition agreement does not extend to that assessment or to the site-specific supplement. Clause 7.5 makes the engaging department responsible for verifying both before a card is issued. The standard sets the requirement correctly. The gap was in the access card system, which had no field for either item and did not distinguish a person who could drive in an operating pit from a person who could not. The verification the standard required had nowhere to take place.

The light vehicle driver's card was issued on 09/08/2026 after a 20 minute gatehouse briefing. His training file showed the Hallorn online module of 24/07/2026 and no pit driving assessment. It was not established whether he had been issued the site-specific supplement, because Hallorn's records for it were not available.

The issue extends beyond this occurrence. Any contractor driver carded through a recognition agreement could enter an operating pit without the site's instruction on how its traffic rules work in practice. That instruction covers giving way to haul trucks and approaching an intersection the driver cannot see out of. Here, the instruction the assessment would have added was to stop at such a mouth and look again, rather than slow. The absence of the check likely contributed to the collision, although no copy of the assessment's content was available.

No safety action has been recorded for this issue.

Positive calls required only at listed intersections, added only at the annual review

Under BIN-TMP-004, a positive call on the pit channel was required only before entering one of the nine intersections listed at Appendix B. An intersection could reach that list only at the plan's annual review. Nothing in the plan required a new road to be referred to its owner when it was cut. The RL 312 spur was not listed, and the absence of a requirement to call there very likely contributed to the occurrence.

The positive call was the mine's own defence against this kind of interaction between vehicles. At this intersection the plan's own terms left that defence without effect. The problem was not confined to the spur. Eleven of the fourteen new Stage 4 roads cut since the last review had never been referred to the plan's owner. No out-of-cycle review had been held in the plan's eleven year history. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision.

Safety action taken

At 05:10 on 12/08/2026, two hours twenty-three minutes after the collision, the Mining Manager of Orrabinda Resources issued a site-wide direction. It required a positive call on the pit channel before entering any intersection in the Stage 4 pit, whether listed in the traffic management plan or not, until further notice. The direction was read at the following four pre-starts and posted beside the Appendix B board in the Stage 4 crib room. It stands until BIN-TMP-004 is amended.

Proposed safety action

On 01/09/2026 the investigation recommended two changes to BIN-TMP-004. The first was a route by which an intersection can reach Appendix B between annual reviews. The second was a requirement that a new road be referred to the plan's owner when it is opened to traffic. A response was due by 30/09/2026, and the recommendation remains open.

Collision alerts suppressed at the spur by a geofence operators knew nothing of

The collision awareness system on the Stage 4 haul fleet raised no alert for the light vehicle at the RL 312 intersection. On 09/06/2026 the contractor's technical services group had extended the S4-LOAD suppression geofence up the main ramp as a system configuration task. Its new boundary covered the place where the spur was cut seven weeks later. It is extremely likely that alerts were suppressed there without operators knowing. It is likely that this contributed by reinforcing the haul truck operator's expectation that a quiet system meant an empty ramp. An unsuppressed advisory alert would have sounded only about half a second after the beacon came into view. The effect therefore lay in the operator's expectation rather than in the alert itself.

The issue extends beyond this intersection. In the twelve months before the occurrence, 68 proximity events inside the geofence were suppressed across the fleet. Nine of them involved a light vehicle on the main ramp below the boundary, all after 09/06/2026. None had been reported to anybody. No training on site, for any person, covered the geofences.

Safety action by Tarrowen Mining Services

On 28/08/2026, Tarrowen technical services added suppressed proximity events to the monthly collision awareness report, broken down by geofence and by vehicle class. It also re-cut the previous twelve months on the same basis for the Stage 4 fleet. This action was in progress, with the first report due 30/09/2026.

Proposed safety action

The investigation recommended that a change to a suppression geofence boundary be treated as a change to a documented control under ORR-PRO-022 clause 5.2. It also recommended that operators be told in writing where the collision awareness system is suppressed and why. This recommendation was open, with a response due 30/09/2026.

No stop control or warning sign at an intersection failing the sight distance rule

The RL 312 intersection had no stop control, give-way control, delineation, warning sign or posted speed, even though its sight distance failed BIN-TMP-004 clause 6.2. It is virtually certain that these controls were absent. With 41 m available against a stopping distance of 96 m, clause 6.2 required a stop control, signalised control or traffic controller. The clause 6.4 signs and give-way control applied only to intersections shown on the road layout drawing, and this intersection was not on it. The construction sign at the spur mouth was removed on 02/08/2026, when the spur was first used, and nothing replaced it. Hazard report HAZ-2026-0917 of 04/08/2026 named the missing sign and the missing give-way.

The absence of these controls likely contributed to the collision. A stop control would have held the light vehicle at the mouth while the truck came into view. A warning sign would have told the haul truck operator that there was an intersection below the crest. The margin from a stop with 38 m of view was thin, however, and the effect of a sign on the truck rests on the operator's own account.

Safety action

The Mining Manager, Orrabinda Resources, recorded that the RL 312 spur was barricaded and closed to all traffic on the night of the occurrence, and it remains closed. Survey then measured the available sight distance at the other ten unlisted roads cut in Stage 4 since the last review, ahead of the November reissue of Schedule 1. Two of the ten were below the clause 6.2 minimum

and have been closed pending a stop control. The action is closed for the RL 312 spur and remains open for those two roads.

Haul operators rostered for fourteen consecutive 12.5 hour night shifts

The contractor's Stage 4 haul roster placed operators on fourteen consecutive night shifts of 12 hours 30 minutes, from 18:00 to 06:30 local with a 30 minute crib. That sequence was followed by 7 nights off. The roster sat within the contractor's fatigue management standard, which permits up to 14 consecutive night shifts with a minimum 10 hour break. The night shift supervisor works the same pattern as the crews he supervises.

On the night of the occurrence, the haul truck operator was on night 10 of 14. The collision happened 8 hours 47 minutes into the shift, inside the window of lowest circadian alertness. Every limit in the standard had been met. The roster was not found to have contributed to this collision, because the operator's recorded response did not show the slowed or absent reaction through which it could have acted.

The roster likely increased risk all the same. It is a standing exposure built into every swing rather than a feature of one night. On each swing, every haul truck operator drives a 220 tonne vehicle through the small hours of a tenth, twelfth and fourteenth consecutive night. At an unlisted intersection, the operator's own detection and response is then the defence against an unlit hazard. Where the sight line is longer than at RL 312, a slower response could decide whether a truck stops in time.

No safety action has been recorded in response to this safety issue.

Night shift extrication waits on a second responder called from the village

It is virtually certain that extrication on night shift at Bindarra could not begin until a second responder arrived from the accommodation village with the hydraulic rescue tool.

The night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder was called in from the village, 11 km from the gatehouse. They travelled in the second response vehicle, which was garaged there and carried the tool.

On the night of the collision, the paramedic's primary survey was complete and the extrication plan made at 02:58. The second responder arrived with the tool at 03:07. The driver was extricated at 03:14, 27 minutes after the collision. The post-incident review measured the delay at approximately 9 minutes. It found that the delay did not affect the clinical outcome, because the injuries were stable and the driver was conscious throughout. The collision and the injuries were complete before the team was paged.

The arrangement very likely increased risk all the same. On this occasion the driver was conscious, restrained and protected by an intact roll-over structure. The same wait would have mattered far more with a fuel fire, an unconscious casualty, or an airway that needed the vehicle opened in the first ten minutes. On every night shift, the site's extrication capability rested on one person plus a drive from the village. Nothing in the arrangement changed with the severity of the casualty it was called to.

No safety action has been recorded in response to this safety issue.

Distribution

Title	Name
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Fictional case study for training. Not a real investigation.

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